

PAPER_1405 — UQFF Derivation of Solar Dynamo Hale Cycle (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** G — Astrophysics (CLOSED — EXACT) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — EXACT integer-primitive identity **Calculator surface:** `calculate_astrophysics({'system': 'solar_dynamo'})`
→ $D_{\text{crit}} - D_{\text{phys}} = 22 \text{ yr}$ EXACT

Observation

Sun's magnetic-polarity reversal cycle is observed at 22 years (Hale 1908). MHD dynamo models reproduce ~22 yr through parameter tuning. UQFF derives it as an integer-primitive identity.

UQFF Closed Identity

$T_{\text{Hale}} = D_{\text{crit}} - D_{\text{phys}} = 26 - 4 = 22 \text{ years}$ EXACT

Physical Interpretation

The bosonic-string critical dimension ($D_{\text{crit}} = 26$) minus spacetime dimensions ($D_{\text{phys}} = 4$) sets the Sun's structural magnetic-cycle period.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
 - Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).
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